

2009 MJC H2 MATHS (9740) JC 2 PRELIM EXAM P1 – SOLUTIONS

Qn	Solution
1	MOD
	$\frac{2}{r^2 + 6r + 8} = \frac{1}{r + 2} - \frac{1}{r + 4}$ $\sum_{r=1}^n \frac{2}{r^2 + 6r + 8} = \sum_{r=1}^n \left(\frac{1}{r + 2} - \frac{1}{r + 4} \right)$ $= \frac{1}{3} - \frac{1}{5}$ $+ \frac{1}{4} - \frac{1}{6}$ $\vdots$ $+ \frac{1}{n+1} - \frac{1}{n+3}$ $+ \frac{1}{n+2} - \frac{1}{n+4}$ $= \frac{1}{3} + \frac{1}{4} - \frac{1}{n+3} - \frac{1}{n+4}$ $= \frac{7}{12} - \left(\frac{1}{n+3} + \frac{1}{n+4} \right)$ $\sum_{r=1}^n \frac{r^2 + 6r + 88}{r^2 + 6r + 8} = \sum_{r=1}^n \left(1 + \frac{80}{r^2 + 6r + 8} \right)$ $= n + 40 \sum_{r=1}^n \frac{2}{r^2 + 6r + 8}$ $= n + 40 \left[\frac{7}{12} - \left(\frac{1}{n+3} + \frac{1}{n+4} \right) \right]$ $= n + \frac{70}{3} - 40 \left(\frac{1}{n+3} + \frac{1}{n+4} \right)$
2	Mathematical Induction – Recurrence Relation
	Let P_n be the statement $S_n = 1 - \frac{1}{x_{n+1} - 2}$ for all $n \in \mathbb{Z}^+$. To prove P_1 is true: $\text{LHS: } S_1 = \sum_{i=1}^1 \frac{1}{x_i - 1}$ $= \frac{1}{x_1 - 1} = \frac{1}{2}$

$$\begin{aligned}
&\text{RHS: } 1 - \frac{1}{x_2 - 2} \\
&= 1 - \frac{1}{x_1^2 - 3x_1 + 4 - 2} \quad (\text{sub in } x_1 = 3) \\
&= 1 - \frac{1}{2} = \frac{1}{2} = \text{LHS}
\end{aligned}$$

$\therefore P_1$ is true.

Assume P_k is true for some $k \in \mathbb{Z}^+$, i.e. $S_k = 1 - \frac{1}{x_{k+1} - 2}$.

To prove P_{k+1} is also true, i.e. $S_{k+1} = 1 - \frac{1}{x_{k+2} - 2}$.

$$\begin{aligned}
\text{LHS: } S_{k+1} &= \sum_{i=1}^{k+1} \frac{1}{x_i - 1} \\
&= \sum_{i=1}^k \frac{1}{x_i - 1} + \frac{1}{x_{k+1} - 1} \\
&= 1 - \frac{1}{x_{k+1} - 2} + \frac{1}{x_{k+1} - 1} \\
&= 1 - \left(\frac{1}{x_{k+1} - 2} - \frac{1}{x_{k+1} - 1} \right) \\
&= 1 - \left[\frac{x_{k+1} - 1 - x_{k+1} + 2}{(x_{k+1} - 2)(x_{k+1} - 1)} \right] \\
&= 1 - \left(\frac{1}{x_{k+1}^2 - 3x_{k+1} + 2} \right) \\
&= 1 - \left(\frac{1}{x_{k+1}^2 - 3x_{k+1} + 4 - 2} \right) \\
&= 1 - \left(\frac{1}{x_{k+2} - 2} \right) = \text{RHS.}
\end{aligned}$$

	$\therefore P_k$ is true $\Rightarrow P_{k+1}$ is also true. Since P_1 is true and P_k is true $\Rightarrow P_{k+1}$ is also true, by Mathematical Induction, P_n is true for all $n \in \mathbb{Z}^+$.
Q3	Integration by substitution
	$\int \frac{\ln x - \ln 2}{x\sqrt{\ln x - \ln 2 - 2}} dx$ $= \int \frac{\ln 2e^t - \ln 2}{2e^t \sqrt{\ln 2e^t - \ln 2 - 2}} \frac{dx}{dt} dt$ $= \int \frac{\ln 2 + t - \ln 2}{2e^t \sqrt{\ln 2 + t - \ln 2 - 2}} 2e^t dt$ $= \int \frac{t}{\sqrt{t-2}} dt$ $= \int \frac{(t-2)+2}{\sqrt{t-2}} dt$ $= \int (t-2)^{\frac{1}{2}} + \frac{2}{\sqrt{t-2}} dt \quad (\text{shown})$ $\int_{2e^2}^{2e^4} \frac{\ln x - \ln 2}{x\sqrt{\ln x - \ln 2 - 2}} dx$ $= \int_2^4 (t-2)^{\frac{1}{2}} + \frac{2}{\sqrt{t-2}} dt$ $= \left[\frac{2}{3}(t-2)^{\frac{3}{2}} + 4(t-2)^{\frac{1}{2}} \right]_2^4$ $= \left[\frac{2}{3}(4-2)^{\frac{3}{2}} + 4(4-2)^{\frac{1}{2}} \right]$ $= \frac{2}{3}(2)^{\frac{3}{2}} + 4(2)^{\frac{1}{2}}$ $= (2)^{\frac{1}{2}} \left[\frac{2}{3}(2) + 4 \right]$ $= \frac{16\sqrt{2}}{3}$ Therefore $a = 16$, $b = 3$
Q4	DE

Let B denote the number of births and D denote the number of deaths.

$$\text{Given } \frac{dB}{dt} \propto x \Rightarrow \frac{dB}{dt} = ax$$

$$\text{and } \frac{dD}{dt} \propto x^2 \Rightarrow \frac{dD}{dt} = bx^2 \quad \text{where } a, b \text{ are positive constants}$$

$$\frac{dx}{dt} = \frac{dB}{dt} - \frac{dD}{dt}$$

$$\therefore \frac{dx}{dt} = ax - bx^2 \quad (\text{shown})$$

$$\text{Given that } \frac{dx}{dt} = 0 \text{ when } x = 200,$$

$$\Rightarrow 200a - 200^2b = 0$$

$$\Rightarrow a = 200b$$

$$\therefore \frac{dx}{dt} = ax - bx^2 = bx(200 - x)$$

$$\int \frac{1}{x(200 - x)} dx = b \int dt$$

$$\int \frac{1}{200} \left[\frac{1}{x} + \frac{1}{(200 - x)} \right] dx = bt + d$$

$$\frac{1}{200} [\ln(x) - \ln(200 - x)] = bt + d$$

$$\frac{1}{200} \ln \left(\frac{x}{200 - x} \right) = bt + d$$

$$\ln \left| \frac{x}{200 - x} \right| = kt + r$$

$$\left| \frac{x}{200 - x} \right| = e^{kt+r}$$

$$\frac{x}{200 - x} = Ae^{kt}, A = \pm e^r$$

$$\frac{200 - x}{x} = Be^{-kt}$$

$$x = \frac{200}{Be^{-kt} + 1}$$

	As $t \rightarrow \infty$, $x = \frac{200}{Be^{-kt} + 1} \rightarrow 200$ Hence there is a limit to the size of the population. (shown) 200 is the maximum number of ToTo bears the environment can support.
Q5	Vectors – Exclude Planes and Lines
(i)	Since $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$ and A, B and C are collinear, $\therefore k = 2$ $\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC}$ $= \mathbf{b} + 2\mathbf{b} - 2\mathbf{a}$ $= 3\mathbf{b} - 2\mathbf{a}$
(ii)	$\frac{3}{4} \mathbf{a} = \left \mathbf{b} \cdot \frac{\mathbf{a}}{ \mathbf{a} } \right$ $\frac{3}{4} \mathbf{a} ^2 = 4$ $\mathbf{a} ^2 = \frac{16}{3}$ $\mathbf{a} = \frac{4}{\sqrt{3}} = \frac{4\sqrt{3}}{3}$
(iii)	Alternatively, $\frac{3}{4}\mathbf{a} = \left(\mathbf{b} \cdot \hat{\mathbf{a}} \right) \hat{\mathbf{a}} = \frac{(\mathbf{b} \cdot \mathbf{a})(\mathbf{a})}{ \mathbf{a} ^2}$ $\Rightarrow \frac{3}{4} \mathbf{a} = \frac{4 \mathbf{a} }{ \mathbf{a} ^2}$ $\frac{3}{4} \mathbf{a} ^2 = 4$ $\mathbf{a} ^2 = \frac{16}{3}$ $\mathbf{a} = \frac{4}{\sqrt{3}} = \frac{4\sqrt{3}}{3}$

(iv)

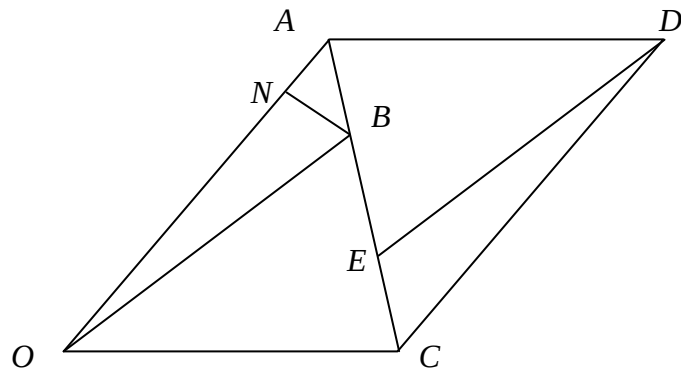
$$|\mathbf{b}| = \sqrt{2^2 + 2^2 + 2^2} = 2\sqrt{3}$$

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}$$

$$= \frac{4}{\left(\frac{4\sqrt{3}}{3}\right)(2\sqrt{3})}$$

$$= \frac{1}{2}$$

$$\theta = \frac{\pi}{3} \text{ or } 60^\circ$$



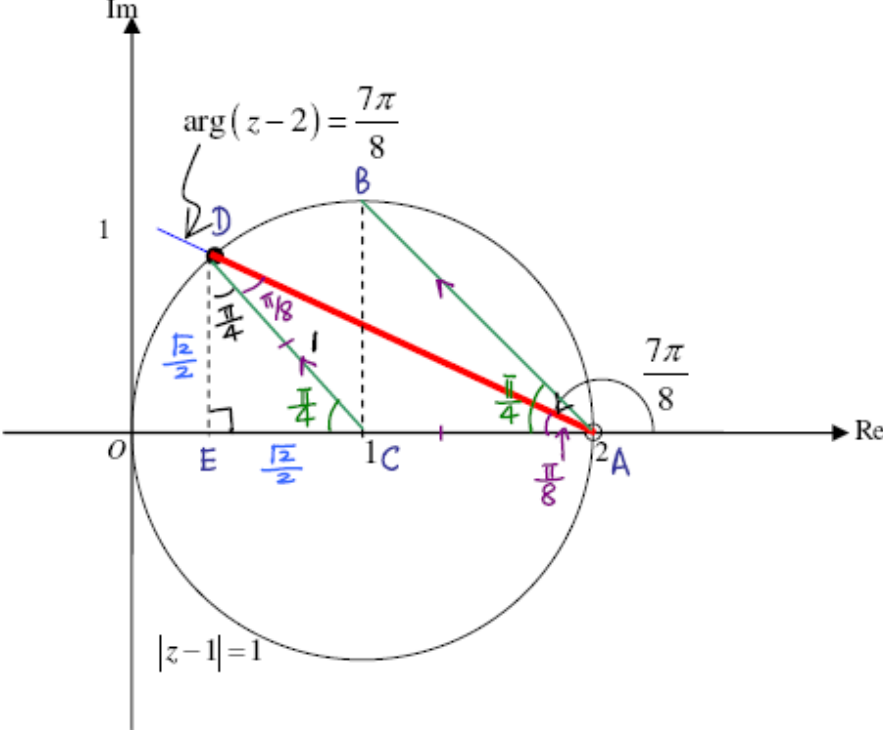
$$\begin{aligned}\overrightarrow{DE} &= \frac{2\overrightarrow{DC} + \overrightarrow{DA}}{3} \\ &= \frac{2(-\mathbf{a}) + (2\mathbf{a} - 3\mathbf{b})}{3}\end{aligned}$$

$$= -\mathbf{b}$$

$$= \begin{pmatrix} -2 \\ 2 \\ -2 \end{pmatrix}$$

Alternatively,

	$\overline{AE} = \frac{2}{3} \overline{AE} = \frac{2}{3}(3\mathbf{b} - 2\mathbf{a} - \mathbf{a}) = 2\mathbf{b} - 2\mathbf{a}$ $\overline{DE} = \overline{DA} + \overline{AE} = \overline{CO} + \overline{AE}$ $= 2\mathbf{a} - 3\mathbf{b} + 2\mathbf{b} - 2\mathbf{a}$ $= -\mathbf{b}$ $= \begin{pmatrix} -2 \\ 2 \\ -2 \end{pmatrix}$
--	---

Q6	Complex Numbers – Loci
(i)	 Diagram illustrating a complex plane with a circle centered at $C(1,0)$ with radius 1, defined by the locus $z-1 =1$. Point A is at $(2,0)$ on the real axis. Point D is on the circle in the first quadrant. A red line segment connects A and D. A blue line segment extends from D. Point E is the projection of D onto the real axis. Point B is at $(1,1)$ on the circle. Various angles are marked: $\angle DCE = \frac{\pi}{4}$, $\angle DAE = \frac{\pi}{8}$, and $\angle DCE = 2\angle DAE$. The argument of $(z-2)$ is given as $\frac{7\pi}{8}$. The circle equation $z-1 =1$ is written at the bottom.
(ii)	$\angle DCE = 2\angle DAE = \frac{\pi}{4}$ (Angle at centre of circle is twice angle at circumference) $\therefore \cos \frac{\pi}{4} = \frac{EC}{1} \Rightarrow EC = \frac{\sqrt{2}}{2}$ $DE = EC = \frac{\sqrt{2}}{2}$

$$\therefore D\left(1-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$

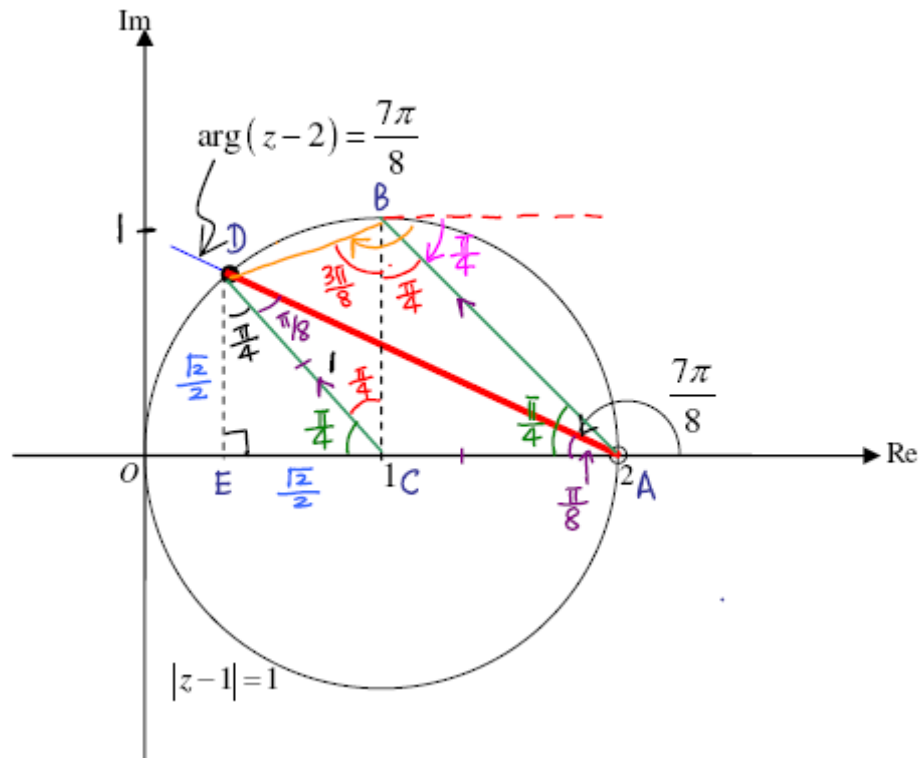
$$z = 1 - \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$$

(iii)

$$\text{Since } \angle EDA = \frac{3\pi}{8},$$

$$\tan \frac{3\pi}{8} = \frac{\frac{\sqrt{2}}{2} + 1}{\frac{\sqrt{2}}{2}} = \sqrt{2} + 1$$

(iv)



Consider Triangle BCD

$$\text{Since } \angle BCD = \frac{\pi}{4},$$

$$\angle DBC = \frac{3\pi}{8} \quad (\text{Isocles Triangle})$$

$$\text{min value of } \arg(z-1-i) = -\frac{\pi}{2} - \frac{3\pi}{8} = -\frac{7\pi}{8}$$

Q7

Complex Numbers – Exclude Loci

(a)	$(1-4i)^2 = -15-8i$ $\left(\frac{z}{2}+3\right)^2 = (-1)(-15-8i) = i^2(1-4i)^2$ $\left(\frac{z}{2}+3\right)^2 = (4+i)^2$ $\frac{z}{2}+3 = 4+i \quad \text{or} \quad \frac{z}{2}+3 = -4-i$ $z = 2+2i \quad \text{or} \quad z = -14-2i$
(b)	$(z-2)^3 = 8 = 2^3 e^{i2k\pi}$ $z-2 = 2e^{i\frac{2k\pi}{3}}$ $z = 2 + 2e^{i\frac{2k\pi}{3}}$ $z = 2\left(e^{i0} + e^{i\frac{2k\pi}{3}}\right)$ $z = 2e^{i\frac{k\pi}{3}}\left(e^{-i\frac{k\pi}{3}} + e^{i\frac{k\pi}{3}}\right)$ $z = 2e^{i\frac{k\pi}{3}}\left(2\cos\frac{k\pi}{3}\right)$ $z = \left(4\cos\frac{k\pi}{3}\right)e^{i\frac{k\pi}{3}}, \quad k = -1, 0, 1$ $z = 2e^{-i\frac{\pi}{3}}, 4 \text{ or } 2e^{i\frac{\pi}{3}}$ $\left(\frac{1-2z}{z}\right)^3 = 8$ $\Rightarrow \left(\frac{1}{z}-2\right)^3 = 8, \quad z \neq 0$ $\Rightarrow z = \frac{1}{2}e^{i\frac{\pi}{3}}, \frac{1}{4} \text{ or } \frac{1}{2}e^{-i\frac{\pi}{3}}$
Q8	Binomial Expansion and Maclaurin's Series
(a)	Method 1 Let $f(x) = (1+e^x)^n$. $f'(x) = n(1+e^x)^{n-1}(e^x)$ $f''(x) = n(n-1)(1+e^x)^{n-2}(e^x)^2 + n(1+e^x)^{n-1}(e^x)$

Therefore, $f(0) = 2^n$; $f'(0) = n(2)^{n-1}$ and $f''(0) = n(n-1)(2)^{n-2} + n(2)^{n-1} = n(n+1)(2)^{n-2}$

$$f(x) = 2^n + n2^{n-1}x + \frac{[n(n+1)2^{n-2}]}{2!}x^2 + \dots$$

$$\approx 2^n + n2^{n-1}x + [n(n+1)2^{n-3}]x^2$$

Method 2

Using standard series $e^x = 1 + x + \frac{x^2}{2}$ (taking the first three terms only)

$$\left(1 + 1 + x + \frac{x^2}{2}\right)^n = \left(2 + x + \frac{x^2}{2}\right)^n$$

$$= 2^n \left(1 + \frac{x}{2} + \frac{x^2}{4}\right)^n$$

$$= 2^n \left[1 + n\left(\frac{x}{2} + \frac{x^2}{4}\right) + \frac{n(n-1)}{2}\left(\frac{x}{2} + \frac{x^2}{4}\right)^2 + \dots\right]$$

$$= 2^n \left[1 + \frac{n}{2}x + \frac{n}{4}x^2 + \frac{n(n-1)}{2}\left(\frac{x}{2}\right)^2 + \dots\right]$$

$$= 2^n \left[1 + \frac{n}{2}x + \left(\frac{n}{4} + \frac{n(n-1)}{8}\right)x^2 + \dots\right]$$

$$= 2^n \left[1 + \frac{n}{2}x + \left(\frac{n(n+1)}{2^3}\right)x^2 + \dots\right]$$

$$\approx 2^n + n2^{n-1}x + [n(n+1)2^{n-3}]x^2$$

(bi) Method 1

$$y = \frac{x}{3-2x}$$

	$y = (x)(3-2x)^{-1}$ $= \frac{x}{3} \left(1 - \frac{2}{3}x\right)^{-1}$ $= \frac{x}{3} \left(1 + (-1) \left(-\frac{2}{3}x\right) + \frac{(-1)(-2)}{2!} \left(-\frac{2}{3}x\right)^2 + \dots\right)$ $= \frac{x}{3} \left(1 + \frac{2}{3}x + \frac{4}{9}x^2 + \dots\right)$ $= \frac{1}{3}x + \frac{2}{9}x^2 + \frac{4}{27}x^3 + \dots$ Method 2 $y = \frac{x}{(3-2x)}$ $= -\frac{1}{2} + \frac{3}{2} \left(\frac{1}{3-2x}\right)$ $= -\frac{1}{2} + \frac{3}{2} \left(\frac{1}{3-2x}\right)$ $= -\frac{1}{2} + \frac{3}{2} (3-2x)^{-1}$ $= -\frac{1}{2} + \frac{3}{2} \left(\frac{1}{3}\right) \left(1 - \frac{2}{3}x\right)^{-1}$ $= -\frac{1}{2} + \frac{1}{2} \left(1 + (-1) \left(-\frac{2}{3}x\right) + \frac{(-1)(-2)}{2} \left(-\frac{2}{3}x\right)^2 + \frac{(-1)(-2)(-3)}{3} \left(-\frac{2}{3}x\right)^3 + \dots\right)$ $= -\frac{1}{2} + \frac{1}{2} + \frac{1}{3}x + \frac{2}{9}x^2 + \frac{4}{27}x^3 + \dots$ $= \frac{1}{3}x + \frac{2}{9}x^2 + \frac{4}{27}x^3 + \dots$
(bii)	$\left \frac{2}{3}x\right < 1$ Therefore the set of values of $x = \left\{x : x \in \mathbb{R}, -\frac{3}{2} < x < \frac{3}{2}\right\}$

(biii))	$y = \frac{x}{3-2x}$ $= \frac{1}{3}x + \frac{2}{9}x^2 + \frac{4}{27}x^3 + \dots$ Clearly the coefficient of x^n follows a GP with first term $\frac{1}{3}$ and common ratio $\frac{2}{3}$ Therefore coefficient of $x^n = T_n = ar^{n-1} = \frac{1}{3}\left(\frac{2}{3}\right)^{n-1}$ Let $y = f(x)$. $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$ By considering the coefficient of x^n for the Maclaurin's expansion, at $x = 0$, $\frac{f^{(n)}(0)}{n!} = \frac{1}{3}\left(\frac{2}{3}\right)^{n-1}$ $\therefore f^{(n)}(0) = n! \left[\frac{1}{3}\left(\frac{2}{3}\right)^{n-1} \right].$
Q9 Vectors – Planes and Lines	
(i)	Sub $(\alpha, \beta, 0)$ into Π_1 and Π_2. $\pi_1: 1(\alpha) + 3(\beta) + a(0) = 8$ $\pi_2: 3(\alpha) + 1(\beta) + b(0) = 0$ Using GC, $\alpha = -1, \beta = 3$
(ii)	$\begin{pmatrix} 1 \\ 3 \\ a \end{pmatrix} \times \begin{pmatrix} 3 \\ 1 \\ b \end{pmatrix} = \begin{pmatrix} 3b - a \\ 3a - b \\ -8 \end{pmatrix}$ $l_1: r = \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3b - a \\ 3a - b \\ -8 \end{pmatrix}, \lambda \in \mathbb{R}$

(iii)

$$\begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3b-a \\ 3a-b \\ -8 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix}$$

$$\Rightarrow \lambda = -\frac{1}{4}$$

Sub $a = -b$ and $\lambda = -\frac{1}{4}$, we have

$$-1 - \frac{1}{4}(-4a) = 5 + 4\mu$$

$$3 - \frac{1}{4}(4a) = 2 + \mu$$

$$\therefore \quad \begin{aligned} a &= 2 \\ b &= -2 \end{aligned}$$

(iv)

Since the 3 planes have no point in common, l_1 cannot intersect Π_3 .

$$l_1 : r = \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}, \lambda \in \mathbb{R}$$

Condition 1:

$(-1, 3, 0)$ is not on Π_3 .

Hence,

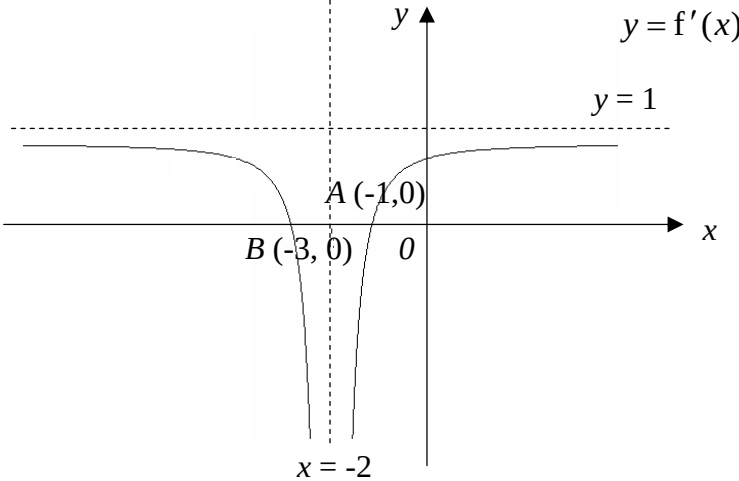
$$(-1) + p(3) + q(0) \neq 11$$

$$\Rightarrow p \neq 4$$

Condition 2:

$\begin{pmatrix} 1 \\ p \\ q \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$ are perpendicular.

	$\Rightarrow \begin{pmatrix} 1 \\ p \\ q \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} = 0$ $\Rightarrow -1 + p - q = 0$ $\Rightarrow p - q = 1$
Q10	AP, GP
(a)	Given $\frac{a(1-r^3)}{1-r} = \frac{26}{27} \left(\frac{a}{1-r} \right) \Rightarrow 1-r^3 = \frac{26}{27} \quad (\because a \neq 0, r \neq 1)$ $\Rightarrow r^3 = \frac{1}{27}$ $\Rightarrow r = \frac{1}{3}$ Since we have $ar^2 - ar = d \Rightarrow ar(r-1) = d$ $\Rightarrow \frac{1}{3}a \left(-\frac{2}{3} \right) = d$ $\Rightarrow -\frac{2}{9}a = d$ Therefore the sum of the first fifty-five terms of the A.P. $= \frac{55}{2} \left[2a + 54 \left(-\frac{2}{9}a \right) \right] = -275a \quad (\text{Ans})$
(b)	Arithmetic Progression with first term, $T_1 = 30$, common difference, $d = 5$, in minutes. (i) $T_7 = T_1 + 6d = 30 + 6(5) = 60$ minutes = 1 hour. (ii) $S_7 = \frac{7}{2}[30 + 60] = 315$ minutes = 5.25 hours (iii) To determine the n^{th} lesson, where Ann would have completed 60 hours (3600 minutes) of lesson. Note n must be a positive integer. $\frac{n}{2}[2(30) + (n-1)(5)] \geq 3600$ $n[55 + 5n] \geq 7200$ $n^2 + 11n - 1440 \geq 0$ $n \geq 32.8 \text{ or } n \leq -43.8$ (Note that students can use GC to get n) Hence during the <u>33rd lesson</u>, Ann will have completed a total of 60 hours.

	Alternatively, $S_n = \frac{n}{2} [2(30) + (n-1)(5)] \geq 3600$ Using GC, when $n = 32$, $S_n = 3440 < 3600$ when $n = 33$, $S_n = 3630 > 3600$ Hence during the <u>33rd lesson</u>, Ann will have completed a total of 60 hours.
Q11	Differentiation (Applications and Graphs of f')
a	 The graph shows the derivative function $y = f'(x)$. It has a vertical asymptote at $x = -2$ and a horizontal asymptote at $y = 1$. The curve passes through the points $A(-1, 0)$ and $B(-3, 0)$. The origin is labeled O.
b	$\frac{dx}{dt} = -a \sin t$; $\frac{dy}{dt} = a \cos t$. $\frac{dy}{dx} = -\cot t$ When $x = \frac{a}{2}$, $t = \frac{\pi}{3}$ and $y = \frac{a\sqrt{3}}{2}$ Equation of tangent: $y - \frac{\sqrt{3}}{2}a = -\frac{\sqrt{3}}{3}\left(x - \frac{a}{2}\right)$ $y = -\frac{\sqrt{3}}{3}x + \frac{\sqrt{3}}{3} \frac{a}{2} + \frac{\sqrt{3}}{2}a$ $= -\frac{\sqrt{3}}{3}x + \frac{\sqrt{3}}{2}a\left(\frac{4}{3}\right)$ $3y = \sqrt{3}(-x + 2a)$

	At $x = 0$: $y = \frac{2\sqrt{3}}{3}a$ At $y = 0$: $x = 2a$ Therefore, area of triangle $OQR = \frac{1}{2}(2a)\left(\frac{2\sqrt{3}}{3}a\right) = \frac{2\sqrt{3}}{3}a^2$								
C	Let the internal height be h, external length be Y and the external height be Z. Volume = 1000 cm^3 $\therefore x^2 h = 1000 \Rightarrow h = \frac{1000}{x^2}$ Also, $Y = 2 + x$ and $Z = 1 + h$ Material needed, $S = Y^2 Z - x^2 h$ $\begin{aligned} &= (2 + x)^2 \left(1 + \frac{1000}{x^2}\right) - x^2 \left(\frac{1000}{x^2}\right) \\ &= x^2 + 4 \left(\frac{1000}{x^2} + 1\right)(x + 1) \end{aligned}$ $\frac{dS}{dx} = 2x + 4(x + 1)\left(-\frac{2000}{x^3}\right) + 4\left(\frac{1000}{x^2} + 1\right) = 0$ $\therefore 2x^4 + 4x^3 - 4000x - 8000 = 0$ $x = -2$ (rejected) or $x = 12.6 \text{ cm}$ <table><tr><td>x</td><td>12.599⁻</td><td>12.599</td><td>12.599⁺</td></tr><tr><td>$\frac{dV}{dx}$</td><td></td><td></td><td></td></tr></table> Answers for non-contextual questions have to be exact when using the test.	x	12.599 ⁻	12.599	12.599 ⁺	$\frac{dV}{dx}$			
x	12.599 ⁻	12.599	12.599 ⁺						
$\frac{dV}{dx}$									